

GPS-Based Ionospheric Tomography From the Combination of PolSAR and E-CHAIM

Cheng Wang¹, Hai-Sheng Zhao², Li-Ming Wang³, Le Cao⁴, *Member, IEEE*,
Zan-Yang Xing⁵, and Yuan-Yuan Zhang⁶

Abstract—The utilization of global positioning system (GPS) for 3-D ionospheric electron density reconstruction, i.e., computerized ionospheric tomography (CIT), provides significant importance in investigating the internal structure, variations, and disturbances within the ionosphere. However, the ill-posed problem caused by insufficient observational data or uneven distribution will rely heavily on the selection of initial values, which are typically derived from empirical models with low precision. Aiming at this issue, this article uses the total electron content (TEC) obtained by the spaceborne full-polarization synthetic aperture radar (PolSAR) to correct the Empirical Canadian High Arctic Ionospheric Model (E-CHAIM), thus improving the authenticity of the initial value. This study makes full use of the advantages of low-frequency full-PolSAR in ionospheric sounding, including high precision and resolution, as well as all-day and all-weather operation without a ground receiver. Therefore, the precision of GPS-based tomography can be enhanced, particularly for small-scale anomalies, and it is also simple and easy to achieve. Numerical and measured experiments using GPS, incoherent scatter radar (ISR), PolSAR, and E-CHAIM data in Alaska demonstrate that the reconstruction accuracy of the proposed CIT is significantly improved than that of the tomography results using only empirical model. In addition, the effects of PolSAR system errors and voxel size on CIT are analyzed to demonstrate the robustness of the method proposed in this article.

Index Terms—Computerized ionospheric tomography (CIT), electron density, full-polarization synthetic aperture radar (PolSAR).

I. INTRODUCTION

THE ionosphere is a highly dynamic region of the upper atmosphere that influences radio wave propagation. Monitoring and research of the ionosphere are essential for

applications such as navigation, space weather forecasting, and high-frequency communication [1], [2]. However, obtaining accurate ionospheric information on a global scale remains an open challenge. With the development of low-frequency spaceborne synthetic aperture radar (SAR) systems in recent years [3], [4], [5], particularly since the launch of the phased array L-band SAR on board the Advanced Land Observing Satellite (ALOS PALSAR) [6], it has become a promising technology of the ionospheric sounding. The ionosphere has dispersive properties that can cause distortion of SAR echo signals [7], [8], [9], such as destroying the reciprocity of the full-polarization SAR (PolSAR) scattering matrix [10], [11], thereby influencing the performance of SAR systems. However, similar to the global positioning system (GPS), although the ionosphere is the main source of errors in GPS positioning measurement, the ionospheric information contained in the echo makes it the most commonly used technology for ionospheric sounding [12], [13], [14], [15]. Compared with traditional ground-based ionospheric measurements, the spaceborne SAR systems have the characteristics of high resolution, all-time, and all-weather. By measuring the change in the full-polarization SAR scattering matrix, high-quality ionospheric information can be derived [16]. In addition, the two-way propagation characteristic of SAR signals using the same antenna for transmission and reception eliminates the need for ground-based receiving stations, making it easier to implement and providing a new technology for ionospheric detection. According to the historical research progress, the relevant studies include 2-D [i.e., Faraday rotation angle (FRA) or total electron content (TEC)], 1-D (i.e., electron density profile), and 3-D (i.e., tomographic imaging) measurements.

The FRA is caused by the rotation of the polarization plane of an electromagnetic wave propagating through a plasma, such as the ionosphere. The rotation angle is proportional to the electron density, so the FRA and TEC values, which represent the 2-D ionospheric structure, can be calculated from the full-PolSAR scattering matrix data. Through mechanisms [17], [18], [19], [20], [21], FRA estimators [22], [23], [24], [25], [26], simulations and actual measurements [16], [27], [28], [29], [30], [31], a number of researchers have confirmed the validity of this theory. On the basis of current SAR system indicators, the Bickel and Bates's estimator based on circular polarization bases are the optimal [31] and serve as the retrieval algorithm for this article. Pi et al. [28] demonstrated that the high-resolution features of SAR combined with the sub-kilometer to kilometer resolution of FRA (depending on

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Cheng Wang is with the Qian Xuesen Laboratory of Space Technology, China Academy of Space Technology, Beijing 100094, China (e-mail: solskjaer2006@126.com).

Hai-Sheng Zhao is with the National Key Laboratory of Electromagnetic Environment, China Research Institute of Radiowave Propagation, Qingdao 266017, China (e-mail: zhaohaisheng213@163.com).

Li-Ming Wang is with the College of Electric Information, Guangxi Minzu University, Nanning 530006, China (e-mail: lwang@gxmzu.edu.cn).

Le Cao is with the School of Electric and Control Engineering, Xi'an University of Science and Technology, Xi'an 710054, China (e-mail: caole@xust.edu.cn).

Zan-Yang Xing is with the School of Space Science and Physics, Shandong University, Weihai 250100, China (e-mail: xingzanyang@sdu.edu.cn).

Yuan-Yuan Zhang is with the School of Physics, Xidian University, Xi'an 710071, China (e-mail: yyzhang1@xidian.edu.cn).

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the specific case) provide significant advantage for detecting TEC enhancements associated with aurora activities, FRA or TEC gradient in the midlatitude trough region, midlatitude traveling ionospheric disturbances, and plasma bubbles at low latitudes. Wang et al. [32] quantified the precision of TEC inversion to be within 1TECU (1TECU = 10^{16} ele/m²), which is substantially higher than GPS precision. Even though the aforementioned studies have verified and demonstrated the benefits of using SAR in ionospheric sounding, it is not possible to completely exploit the potential of full-PolSAR by using SAR radar alone. To accomplish more precise and efficient ionospheric detection, it is necessary to fully integrate other sensors and models. In this field, Wang et al. [32] and Zhu et al. [33] corrected the 1-D electron density profiles of the ionosonde and IRI model, respectively, using PolSAR data and demonstrated the potential for PolSAR integration with other ionospheric data sources.

However, 3-D detection remains the ultimate objective of ionospheric research, as it can help researchers better understand the processes that drive ionospheric dynamics, such as solar activity, Earth's magnetic field and polar outflow that are difficult to observe in 2-D. This is also important for applications such as satellite communication and navigation. The computerized ionospheric tomography (CIT) technology is the key to achieving 3-D ionospheric distribution [34], [35], [36]. The CIT system uses a technique called "radio tomography" which is similar to the medical imaging technique called computed tomography (CT) [34]. In CT, X-rays pass through the body from different angles, and the data obtained is used to create a 3-D image of the internal structure. Similarly, in CIT, radio waves are transmitted from multiple GPS satellites, travel through the ionosphere, and are received by a network of ground-based receivers. The data gathered by the receivers is then used to produce a 3-D image of the ionosphere, i.e., electron density distribution.

Nevertheless, the CIT itself has limitations, such as a limited number of observation stations and viewing angles, resulting in an ill-posed problem whose coefficient matrix is highly ill-conditioned [37], [38]. In this case, some prior knowledge needs to be incorporated to obtain a unique solution. The most commonly used method is to directly introduce empirical model parameters of the ionosphere as prior information (i.e., initial values) to constrain the reconstruction algorithm, and then use multiplicative algebraic reconstruction technique (MART) or other improved methods for iterative solution to obtain a stable electron density solution [39], [40], [41], [42], [43]. However, as the empirical model can only reflect the average variation characteristics of the ionosphere, large errors may occur. Recently, Wang et al. [44] have successfully corrected the initial values using the electron density values observed by a combination of the PALSAR data and ionosonde data, substantially enhancing the precision of tomographic imaging. Unfortunately, this technique also has limitations: first, it still requires support from ionosondes located on the ground. After interpolation processing of the corrected initial value, the anomaly in the reconstruction area may not be accurately observed if there are few ionosondes. Increasing the quantity of ionosondes will increase the cost. Second, the

observation time requirements are strict. The PALSAR and ionosonde observation data must be collected at the same time and location, and the GPS data observation time also needs to be considered. Therefore, in order to reduce the difficulty of implementing CIT caused by ionosonde, this article will use PALSAR data to correct the Empirical Canadian High Arctic Ionospheric Model (E-CHAIM) model to enhance the accuracy of the iterative initial value. The advantages of this method are: first, there is no longer a need to consider ionosonde, which reduces one type of detection equipment and makes tomography easier to implement; second, there is no need to consider the matching of PALSAR and ionosonde time and location, and only the observation time, location information, and retrieval value of PALSAR need to be inputted into the E-CHAIM model to complete initial value correction; then, as the PALSAR and ECHAIM are not constrained by ground receivers, CIT can theoretically be performed as long as there are GPS receiver stations; finally, combined with the high resolution of SAR, this method may detect small-scale ionospheric anomalies.

The outline of this article is given as follows. In Section II, the basics of ionospheric retrieval using PALSAR, corrections of electron density using PALSAR and E-CHAIM, and CIT proposed in this article are briefly introduced. In Section III, by using the PALSAR full-pol data, the geomagnetic field derived from the International Geomagnetic Reference Field (IGRF) model, the ECHAIM model, and GPS data, simulation and measured results show the feasibility of the proposed technique. Finally, our conclusions are put in Section IV.

II. METHODOLOGY

A. FRA Estimator

The phenomenon of Faraday rotation occurs when polarized electromagnetic waves propagate through a magnetic medium, e.g., the ionosphere. It causes a rotation of the wave's polarization plane, which is quantified by the FRA and can be defined as [28]

$$\Omega = \frac{2.365 \times 10^4 \overline{B \cos \theta_B} \text{TEC}}{f^2} \quad (1)$$

where Ω , B , and θ_B are the FRA, magnitude of the ambient magnetic field, and angle between the radio wave's vector and the magnetic field, respectively. The $\overline{B \cos \theta_B}$ corresponds to the path's mean value of $B \cos \theta_B$.

It can be seen that the FRA is a frequency-dependent physical quantity. In most SAR cases, f varies from $f_0 - B_0/2$ to $f_0 + B_0/2$, where f_0 and B_0 are the carrier frequency and bandwidth, respectively. Consequently, the variation of FRA with f may affect the scattering matrix and the imaging quality of each channel, thereby affecting the accuracy of FRA estimator, i.e., polarization dispersion [46]. However, for the PALSAR system, the relative bandwidth is not large, and the current FRA estimators do not account for the influence of polarization dispersion, i.e., the f is approximated as the f_0 . In order to analyze the effect of SAR on the precision of CIT as realistically as feasible, polarization dispersion will be taken into account in this article.

When FRA is considered, the measured scattering matrix $\mathbf{M}$ can be expressed as

$$\begin{bmatrix} M_{hh} & M_{vh} \\ M_{hv} & M_{vv} \end{bmatrix} = \begin{bmatrix} \cos \Omega(f) & \sin \Omega(f) \\ -\sin \Omega(f) & \cos \Omega(f) \end{bmatrix} \cdot \begin{bmatrix} S_{hh} & S_{vh} \\ S_{hv} & S_{vv} \end{bmatrix} \cdot \begin{bmatrix} \cos \Omega(f) & \sin \Omega(f) \\ -\sin \Omega(f) & \cos \Omega(f) \end{bmatrix} \quad (2)$$

where $\mathbf{S}$ is the scattering matrix. The subscript h and v represent linear horizontal and vertical polarizations, respectively. The FRA can therefore be derived from the above scattering matrix or the corresponding covariance matrix [22], [23], [24], [25], [26]. Among these FRA estimators for the PALSAR system parameters, the measurement method based on the circular polarization scattering matrix is the most commonly used and optimal algorithm [22]. The CIT study in this article also gives priority to using this estimator. For the scattering matrix $\mathbf{Z}$ in circular basis, it can be converted by unitary matrix of incident wave and scattered wave, i.e., [45]

$$\begin{bmatrix} Z_{ll} & Z_{lr} \\ Z_{rl} & Z_{rr} \end{bmatrix} = [\mathbf{U}_D^{C,s}] \mathbf{M} [\mathbf{U}_C^{D,i}] \quad (3)$$

where the subscript l and r represent left- and right-circular polarizations, respectively, and

$$\mathbf{U}_D^{C,s} = \frac{1}{\sqrt{2}} \begin{pmatrix} -j & 1 \\ 1 & -j \end{pmatrix}, \quad \mathbf{U}_C^{D,i} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & j \\ j & 1 \end{pmatrix} \quad (4)$$

where $j = (-1)^{1/2}$. As discussed in [46], the frequency-dependent FRA has no effect on co-polarization channels, but it introduces extra time-domain phase errors in cross-polarization channels

$$\phi_{PD} = \frac{\Omega(f_0) B_0^2}{2f_0^2}. \quad (5)$$

Ignoring the constant term, (3) can be finally expressed as

$$\begin{cases} Z_{ll} = S_{hh} - S_{vv} + 2jS_{hv} \\ Z_{lr} = (S_{hh} + S_{vv}) \exp\left(j\frac{\pi}{2} - j2\Omega(f_0) - j\phi_{PD}\right) \\ Z_{rl} = (S_{hh} + S_{vv}) \exp\left(j\frac{\pi}{2} + j2\Omega(f_0) + j\phi_{PD}\right) \\ Z_{rr} = S_{vv} - S_{hh} + 2jS_{hv} \end{cases} \quad (6)$$

The FRA based on circular basis can therefore be obtained by

$$\Omega' = -\frac{1}{4} \arg(Z_{lr} \cdot Z_{rl}^*) = \Omega(f_0) \left(1 + \frac{B_0^2}{4f_0^2}\right). \quad (7)$$

The FRA at f_0 is then derived as

$$\Omega(f_0) = \frac{4f_0^2 \Omega'}{4f_0^2 + B_0^2}. \quad (8)$$

Thus, the slant TEC is further calculated according to (1) and (8)

$$\text{TEC}_{\text{SAR}} = \frac{f_0^2 \Omega(f_0)}{2.365 \cdot 10^4 \langle B \cos \theta_B \rangle}. \quad (9)$$

Based on the single-layer model mapping function (SLM-MF) [47] and satellite ground range geometry [48], the vertical TEC at the height of ionospheric pierce point (IPP) is

$$\text{VTEC}_{\text{SAR}} = \text{TEC}_{\text{SAR}} \cdot \cos \left[\arcsin \left(\frac{\text{Re} + H_s}{\text{Re} + H_{\text{ipp}}} \sin(\phi) \right) \right] \quad (10)$$

where $\text{Re} = 6368.137$ km, $H_{\text{ipp}} = 300$ km, $H_s = 691.65$ km, and $\phi = 21.5^\circ$ represent the Earth's radius, the height of IPP, the height of PALSAR, and the off-nadir angle of the PALSAR, respectively. Thus, according to the above parameters, $\text{VTEC}_{\text{SAR}} = 0.92\text{TEC}_{\text{SAR}}$, without considering the curvature of the Earth, $\text{VTEC}_{\text{SAR}} = 0.93\text{TEC}_{\text{SAR}}$.

In practice, a full-PolSAR system still contains residual calibration errors and noise. Taking into consideration these effects, the complete expression of (2) is

$$\begin{bmatrix} M'_{hh} & M'_{vh} \\ M'_{hv} & M'_{vv} \end{bmatrix} = \begin{bmatrix} 1 & \delta_2 \\ \delta_1 & f_1 \end{bmatrix} \cdot \begin{bmatrix} M_{hh} & M_{vh} \\ M_{hv} & M_{vv} \end{bmatrix} \cdot \begin{bmatrix} 1 & \delta_3 \\ \delta_4 & f_2 \end{bmatrix} + \begin{bmatrix} N_{hh} & N_{vh} \\ N_{hv} & N_{vv} \end{bmatrix} \quad (11)$$

where δ_i , $i = 1 \sim 4$ are the crosstalk terms, f_1 and f_2 are the channel imbalance terms and N_{hh} , N_{vh} , N_{hv} , N_{vv} are the additive noise in each channel.

B. E-CHAIM Model Correction Based on PALSAR

CIT is significantly influenced by the initial value selection. The greater the difference between the initial value and the true value, the greater the deviation of the tomographic results from the actual situation. In most cases, iterative initial value can be obtained from empirical ionospheric models such as the International Reference Ionosphere (IRI) model [38], [42], [43] or the E-CHAIM model [49], the latter of which is more appropriate for high northern latitudes and is used in this article. However, the empirical model relies on simplified assumptions and parameterizations to represent the complex and dynamic nature of the ionosphere. Consequently, it may not accurately capture local or temporal variations, especially during geomagnetic disturbances or unusual ionospheric conditions.

As mentioned in Section II-A, since the full-PolSAR provides reliable VTEC values, we can use it to correct the E-CHAIM model and obtain more precise initial information about electron density [33]. The VTEC value obtained from the E-CHAIM model can be expressed as

$$\text{VTEC}_E = \sum_{y=1}^N \text{Ne}(h_y) \Delta h \quad (12)$$

where the whole path is divided into N parts and

$$\Delta h = \frac{H_s - H_b}{N}, \quad h_y = H_b + (y - 1) \Delta h. \quad (13)$$

$\text{Ne}(h_y)$ is the electron density at h_y and H_b is the height at the bottom of the ionosphere. When we obtain VTEC using

full-PolSAR data, the electron density values obtained from E-CHAIM model can be corrected by

$$Ne'(h_y) = Ne(h_y) \cdot \frac{VTEC_{SAR}}{VTEC_E}. \quad (14)$$

Since the electron density near the peak height accounts for the major part of the whole path, (14) will make the correction more noticeable near the peak. As discussed in [43] and [44], the enhancement of the electron density value near the peak is the key to enhancing the initial accuracy, which in turn enhance the tomographic precision.

C. 3-D CIT Based on GPS, PALSAR, and E-CHAIM Model

As an inverse problem, the CIT reconstructs the unknown electron density distribution of the ionosphere in a particular region by measuring the TEC along the radio signal propagation paths between a series of GPS satellites and ground receivers. The received TEC can be expressed as the integrated electron density along the propagation path s

$$TEC_i = \int_s Ne(r) ds = \sum_{j=1}^n a_{ij} x_j \quad (15)$$

where TEC_i represents the slant TEC obtained from the i th ray ($i = 1, \dots, m$). The whole area of interest in the ionosphere is divided into n grid voxels s in which the electron density is constant. a_{ij} represents the intercept of the i th ray in the j th voxel ($j = 1, \dots, n$), and x_j represents the electron density value of the ionospheric voxel to be solved. These variables can be represented in matrix form as follows:

$$\mathbf{d}_{m \times 1} = \mathbf{A}_{m \times n} \cdot \mathbf{X}_{n \times 1} \quad (16)$$

where $\mathbf{d}$ is an m -dimensional column vector composed of TEC observation values, $\mathbf{A}$ is an m -by- n matrix consisting of intercepts of rays in corresponding pixels, and $\mathbf{X}$ is an n -dimensional column vector representing the electron density of the ionospheric space to be reconstructed. In CIT, the MART is commonly used to solve the matrix equation above [35], [37]. The modified electron density values obtained by PALSAR and E-CHAIM are utilized as the initial value $\mathbf{X}^{(0)}$ for iteration in MART. The iteration equation is

$$\mathbf{X}_j^{(k+1)} = \mathbf{X}_j^k \left(\frac{d_i}{\langle \mathbf{X}^k, \mathbf{a}_i \rangle} \right)^{\frac{\lambda_k a_{ij}}{\|\mathbf{a}_i\|}} \quad (17)$$

where $\mathbf{X}_j^{(k+1)}$ and $\mathbf{X}_j^k$ are the electron density of the j th voxel at iteration $k+1$ and k , respectively. $\langle \bullet \rangle$ denotes the inner product. $\|\mathbf{a}_i\|$ represents the summation of the i th path in the ionosphere, and the relaxation factor is $\lambda = 0.8$. Therefore, combining Sections II-A and II-B, the experiment procedure of the CIT proposed in this article, which is based on GPS, PALSAR, and ECHAIM models, can be summarized as follows.

- 1) Determine the reconstruction area and time, and discretize the area in the latitude, longitude, and altitude directions, i.e., voxel division.

- 2) Based on GPS data in h5 format [50], read the coordinates of GPS satellites and ground receiving stations within the reconstruction area and time, which are used to calculate the length of the intercepts of all valid rays in each voxel, i.e., matrix $\mathbf{A}$.
- 3) Using the E-CHAIM model to generate the electron density values of each voxel in the reconstruction area and time as the true values $x_{j,real}$, and then using (15) to obtain the TEC_i value of each ray, which is the matrix $\mathbf{d}$.
- 4) Using the “ceos_leader” and “workreport” files in each PALSAR scene within the reconstruction area, we get a set of PALSAR satellite coordinates, off-nadir angle, and ground observation area center point coordinates, and then calculate the corresponding IPPs’ coordinates. According to (10), the simulated PALSAR data is processed using the true values acquired in the previous step to determine the IPP’s VTEC. Another distinct time is selected from the E-CHAIM model, the output of electron density is regarded as the initial value. Corrections are then made to the electron density in the altitude direction voxels corresponding to the IPPs. The corrected electron density is processed by 2-D quadratic polynomial interpolation at different altitudes, and the corrected initial value for 3-D iteration is then obtained.
- 5) The corrected initial values are used as input to solve the tomography based on (17), the number of iterations is 10.

III. 3-D CIT NUMERICAL EXPERIMENT

The ionospheric tomography imaging technique can be implemented with the voxel-based model method. Nevertheless, as discussed in Section II, this method requires high accuracy of the initial values and it is not enough to rely only on empirical model. This article utilizes the VTEC values derived from the full-polarization PALSAR data to correct the model data, thereby enhancing the authenticity of the initial values. Based on simulation and measured data described in Section II-C, this section will demonstrate the algorithm’s superiority in both the quiet and disturbed cases. In addition, the effects of system errors and different voxel resolutions on CIT are studied.

A. CIT Experiments Under Quiet Condition

Through simulations, this section aims to demonstrate that the proposed CIT method (i.e., GPS-PALSAR-ECHAIM) can produce more accurate reconstruction results than the traditional method (i.e., GPS-ECHAIM).

As shown in Fig. 1, we first establish the reconstruction location in Alaska according to the CIT processing procedure, within a latitude range of 59.5° N to 64.5° N and a longitude range of 145.4° W to 150.4° W, at an altitude ranging from 100 to 600 km. The corresponding voxel scale sets are 0.5°, 0.5°, and 10 km, respectively. This results in a total of $10 \times 10 \times 50$ voxels, as shown by the dashed lines in Fig. 1, which represent the division of the longitude and latitude voxel sizes. Next, the reconstruction time is set as 16:00 UT on May 10, 2011. 32 ground receiver coordinates are then

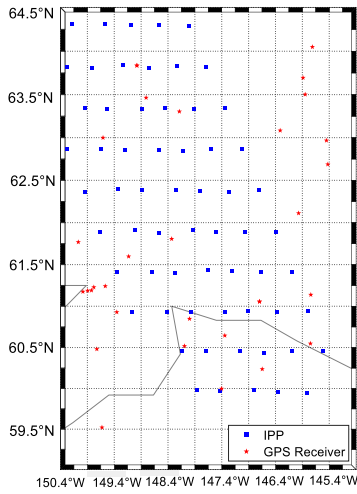


Fig. 1. Geographic relationship of 32 GPS receivers and 63 IPPs of full-polarization PALSAR data.

extracted from the GPS data (in.h5 format) within the reconstruction area, as indicated by the red pentagrams in Fig. 1. In order to verify the performance of the proposed method at different time resolutions, two time resolutions of 10 and 30 min are selected for verification based on ionospheric characteristics [51], [52]. Therefore, the GPS coordinates corresponding to all rays within 5 and 15 min before and after the reconstruction time (i.e., between 15:56 and 16:05 UT on May 10, 2011 and between 15:46 and 16:15 UT on May 10, 2011) are read to calculate the coefficient matrix $\mathbf{A}$. During these two specified time period, 5247 and 15 705 effective rays are obtained from the GPS data, respectively. On the basis of the reconstruction area, time, and voxel parameters, the E-CHAIM model can be utilized to derive the assumed true spatial distribution of electron density, after which the synthetic TEC values $\mathbf{d}$ are calculated using (15). Another time, i.e., 15:30 UT on May 10, 2011 is chosen for E-CHAIM to generate the initial value. This initial value correction will be performed on the 63 scenes of fully polarized PALSAR data in this case. As shown by the blue squares in Fig. 1, the coordinates of 63 IPPs can be calculated using the coordinates of the scene center points (workreport file) and the SAR satellite coordinates (ceos_leader file). The symmetric treatment of the cross-polarized channels assumes an ideal scattering matrix without system errors [26]. By incorporating the real electron density values obtained in the previous step and the geomagnetic field of the IGRF into this ideal condition, the scattering matrix of (2) is obtained. Equation (10) can therefore be used to obtain VTEC_{SAR} , which is the retrieval VTEC in the altitude direction of the voxel where the IPP is located. It should be noted that if a voxel has two or more IPPs, the average VTEC value of both or all IPPs should be taken. Finally, the electron density at 15:30 on May 10, 2011 is selected as the initial value to be corrected. By using (14), the 1-D electron density profiles corresponding to the locations of 63 IPPs are corrected. Then, a horizontal 2-D interpolation process is performed at each altitude, resulting in the corrected 3-D initial electron density values.

In accordance with the aforementioned parameters, Fig. 2 demonstrates the 3-D CIT results using traditional (GPS and E-CHAIM) and proposed (GPS, PALSAR, and E-CHAIM) techniques, respectively. The true distribution is shown in Fig. 2(a). The reconstruction accuracy can be evaluated with the average deviation (AD) and root-mean-square error (RMSE), which are defined as

$$\text{AD} = \text{mean}|x_{j,\text{real}} - x_{j,\text{CIT}}|$$

$$\text{RMSE} = \sqrt{\frac{\sum_{j=1}^n (x_{j,\text{real}} - x_{j,\text{CIT}})^2}{n}} \quad (18)$$

where $x_{j,\text{CIT}}$ is the electron density value of CIT result. In this experiment, the AD and RMSE between the initial value provided only by E-CHAIM and the true value are $1.17 \times 10^{10} \text{ el/m}^3$ and $1.46 \times 10^{10} \text{ el/m}^3$, respectively. After SAR correction, the AD decreases to $6.68 \times 10^9 \text{ el/m}^3$ and the RMSE decreases to $7.42 \times 10^9 \text{ el/m}^3$, indicating a significant improvement in the accuracy of the initial value. As shown in Figs. 2(b) and 3(a), after reconstruction with 10 min GPS data, the AD and RMSE of the CIT using only the empirical model as the initial value are $6.71 \times 10^9 \text{ el/m}^3$ and $9.87 \times 10^9 \text{ el/m}^3$, respectively. Consistent with Fig. 2(b) but using 30-min GPS data, Figs. 2(c) and 3(b) show the reconstructions with an AD of $4.86 \times 10^9 \text{ el/m}^3$ and an RMSE with $7.46 \times 10^9 \text{ el/m}^3$. Figs. 2(d) and 3(c) show the reconstruction using 10-min GPS data based on the proposed method, which the AD and RMSE are $5.52 \times 10^9 \text{ el/m}^3$ and $6.54 \times 10^9 \text{ el/m}^3$, respectively, indicating a more accurate representation of the true distribution. By using 30-min GPS data based on the proposed method, Figs. 2(e) and 3(d) demonstrate the reconstruction results with an AD of $4.06 \times 10^9 \text{ el/m}^3$ and an RMSE of $5.16 \times 10^9 \text{ el/m}^3$. It can be seen that compared with the empirical model, the reconstruction using 10-min GPS data results in a 17.73% decrease in the AD and a 33.74% decrease in the RMSE after SAR correction. Similarly, when utilizing 30 min of GPS data, there is a decrease of 16.46% in the AD and 30.83% in the RMSE. This suggests that under different time resolution conditions, utilizing PALSAR to improve the authenticity of the initial model is a viable method for improving the accuracy of tomographic imaging. In addition, it makes full use of the two-way propagation characteristics of SAR, eliminating the need for ground-based receiving stations and making its implementation simpler.

B. CIT Experiments Under Disturbed Condition

Occasional ionospheric disturbances occur under the influence of factors such as the sun, geomagnetism, earthquake, and atmosphere. In this section, relevant experiments are conducted to validate the algorithm's ability to reconstruct small-scale ionospheric disturbance.

Based on Fig. 2(a), an ionospheric disturbance is added in the range of approximately 62.5° N to 64.5° N , 147° W to 149° W , 220–300 km, as shown in Fig. 4(a), and it is assumed to be the true value of electron density. Other parameters are in accordance with Fig. 2. Compared to Fig. 2, the AD and RMSE between the initial value provided by E-CHAIM and the actual value have increased to $1.38 \times 10^{10} \text{ el/m}^3$ and

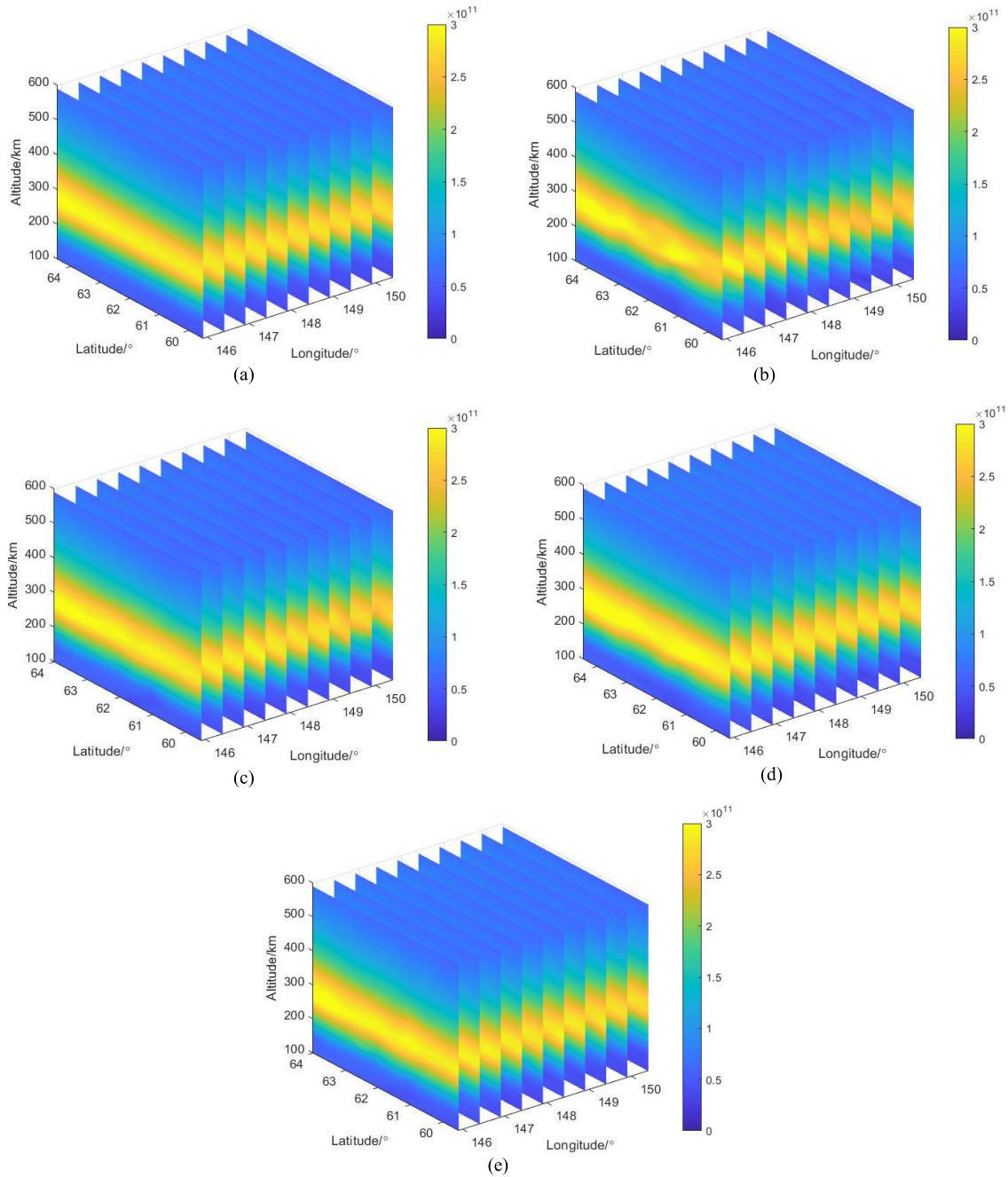


Fig. 2. 3-D CIT reconstructions without system errors. (a) True electron density distribution. (b) and (c) CIT reconstruction result based on GPS and E-CHAIM model with 10 and 30 min, respectively. (d) and (e) CIT reconstruction result based on GPS, PALSAR, and E-CHAIM model with 10 and 30 min, respectively.

$2.01 \times 10^{10} \text{ el/m}^3$, respectively. By using the 10-min GPS data, the CIT's AD and RMSE only using the E-CHAIM model are $9.44 \times 10^9 \text{ el/m}^3$ and $1.67 \times 10^{10} \text{ el/m}^3$, which is shown in Figs. 4(b) and 5(a). Consistent with Fig. 4(b) but using 30-min GPS data, the reconstructed AD and RMSE in Figs. 4(c) and 5(b) are $6.85 \times 10^9 \text{ el/m}^3$ and $1.32 \times 10^{10} \text{ el/m}^3$, respectively. The initial value corrected by PALSAR is still more accurate than the E-CHAIM alone, with an AD of $9.82 \times 10^9 \text{ el/m}^3$ and an RMSE of $1.47 \times 10^{10} \text{ el/m}^3$. As shown in Figs. 4(d) and 5(c), after the proposed CIT using 10-min GPS data, the AD and RMSE are the $8.02 \times 10^9 \text{ el/m}^3$ and $1.23 \times 10^{10} \text{ el/m}^3$, respectively. Similarly, when using the

30-min GPS data, the AD decreases to $5.84 \times 10^9 \text{ el/m}^3$ and RMSE decreases to $1.04 \times 10^{10} \text{ el/m}^3$. Obviously, the proposed method can still increase accuracy compared to the traditional method in the case of disturbance. After reconstruction using 10-min GPS data, the AD of proposed method decreases by 15.04% and the RMSE decreases by 26.51%. Similarly, the reconstruction using 30-min GPS data showed that AD and RMSE decrease by 14.74% and 20.97%, respectively. It should be noted that the key to improving the accuracy of tomography is that there are enough effective rays passing through the disturbed area, and as many vertical grid points as possible in the area are corrected by the PALSAR data, that is, the

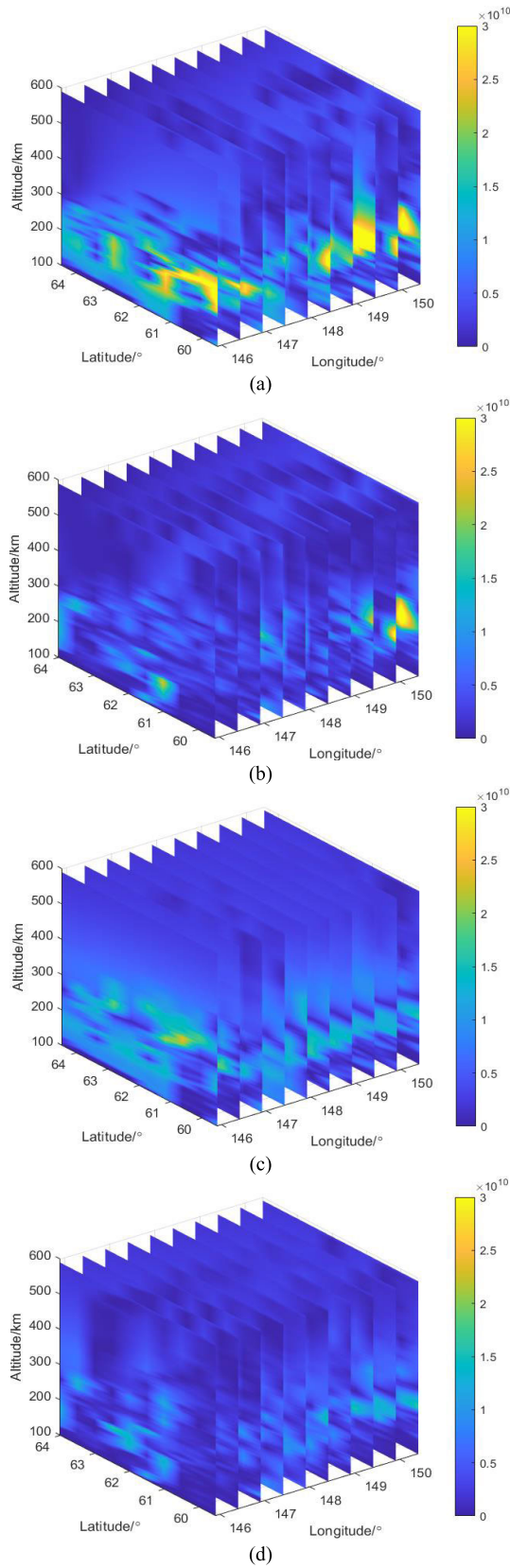


Fig. 3. Deviations of CIT of Fig. 2. (a) and (b) Deviations between the true distribution and reconstructions based on GPS and E-CHAIM with 10 and 30 min, respectively. (c) and (d) Deviations between the true distribution and reconstructions based on GPS, PALSAR, and E-CHAIM with 10 and 30 min, respectively.

number of IPPs in the disturbed area is sufficient. Otherwise, the proposed method may not be effective.

C. CIT Accuracy Under the Condition of System Errors

In practice, after PALSAR system calibration, there are still residual system errors, including noise, channel phase imbalance, channel amplitude imbalance, and crosstalk, and the complete scattering matrix form is shown in (11). Therefore, in order to make the proposed method more realistic, this section analyzes the influence of each system error on the accuracy of the tomography imaging.

When the noise is considered, (11) changes to

$$\begin{aligned} M_{hh}^N &= S_{hh} \cos^2 \Omega(f) - S_{vv} \sin^2 \Omega(f) + N_{hh} \\ M_{vh}^N &= S_{hv} - (S_{hh} + S_{vv}) \sin \Omega(f) \cos \Omega(f) + N_{vh} \\ M_{hv}^N &= S_{hv} + (S_{hh} + S_{vv}) \sin \Omega(f) \cos \Omega(f) + N_{hv} \\ M_{vv}^N &= S_{vv} \cos^2 \Omega(f) - S_{hh} \sin^2 \Omega(f) + N_{vv}. \end{aligned} \quad (19)$$

Assuming that the power of noise in each channel is the same, and the signal-to-noise ratio (SNR) is defined as [26]

$$\text{SNR} = \frac{\langle |S_{hh}|^2 \rangle + \langle |S_{hv}|^2 \rangle + \langle |S_{vh}|^2 \rangle + \langle |S_{vv}|^2 \rangle}{4\langle |N_{hh}|^2 \rangle}. \quad (20)$$

Incorporating the FRA and different levels of noise into the scattering matrix to generate the scattering matrix form of (19), we then perform the processing of (8), (10), and CIT with the parameters consistent with Figs. 2 and 4. Figs. 6(a) and 7(a) illustrate the AD and RMSE of CIT with SNR under two time resolutions, and 100 Monte Carlo simulations are performed on each SNR parameter. Overall, the accuracy increases with the increase of SNR, but it is relatively stable. Similarly, when only the channel imbalance and crosstalk are considered, respectively, the scattering matrix is

$$\begin{aligned} M_{hh}^{f'} &= S_{hh} \cos^2 \Omega(f) - S_{vv} \sin^2 \Omega(f) \\ M_{vh}^{f'} &= f' [S_{hv} - (S_{hh} + S_{vv}) \sin \Omega(f) \cos \Omega(f)] \\ M_{hv}^{f'} &= f' [S_{hv} + (S_{hh} + S_{vv}) \sin \Omega(f) \cos \Omega(f)] \\ M_{vv}^{f'} &= f'^2 (S_{vv} \cos^2 \Omega(f) - S_{hh} \sin^2 \Omega(f)) \end{aligned} \quad (21)$$

and

$$\begin{aligned} M_{hh}^{\delta'} &= M_{hh} + \delta' M_{hv} + \delta' M_{vh} + \delta'^2 M_{vv} \\ M_{vh}^{\delta'} &= \delta' M_{hh} + \delta'^2 M_{hv} + M_{vh} + \delta' M_{vv} \\ M_{hv}^{\delta'} &= \delta' M_{hh} + M_{hv} + \delta'^2 M_{vh} + \delta' M_{vv} \\ M_{vv}^{\delta'} &= \delta'^2 M_{hh} + \delta' M_{hv} + \delta' M_{vh} + M_{vv}. \end{aligned} \quad (22)$$

For convenience, $f' = f_1 = f_2$ and $\delta' = \delta_1 = \delta_2 = \delta_3 = \delta_4$ are assumed. Similarly, Figs. 6(b)–(d) and 7(b)–(d), respectively, show the CIT accuracy with the changes in channel phase imbalance, amplitude imbalance, and crosstalk. These three system errors vary from 1° to 5° , 0.5–1 dB, and -45 to -25 dB [16], respectively. From the results, it is evident that the PALSAR system errors have good robustness for CIT, with a maximum difference of about only $0.2 \times 10^9 \text{ el/m}^3$ between the worst and best conditions. This is due to the fact that, unlike other FRA estimators, Bickel and Bates' estimator is insensitive to system errors and can generally retrieve optimal FRA or TEC values. Other FRA estimators perform poorly in some cases.

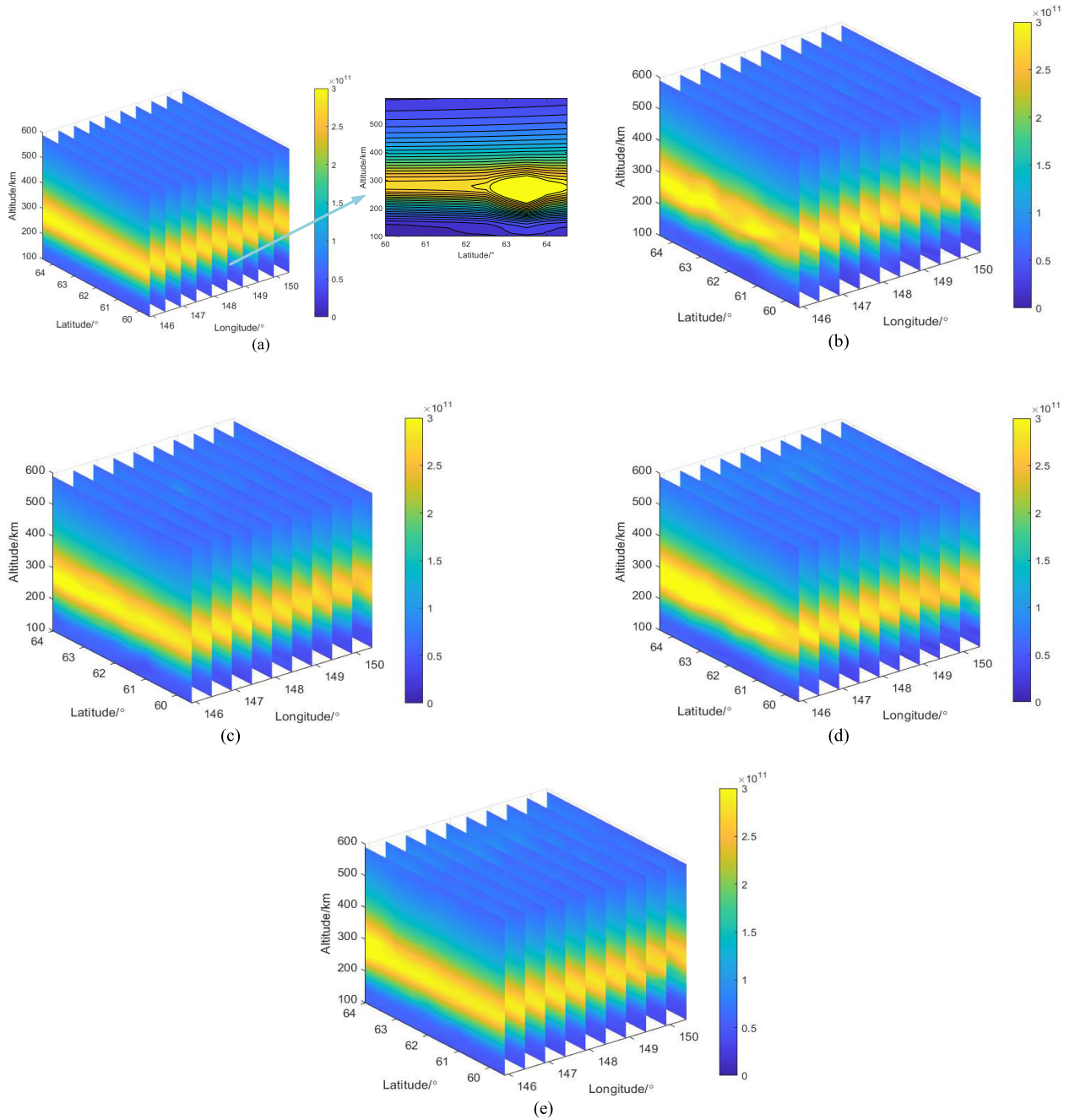


Fig. 4. 3-D CIT reconstructions with disturbance. (a) True electron density distribution. (b) and (c) CIT reconstruction result based on GPS and E-CHAIM model with 10 and 30 min, respectively. (d) and (e) CIT reconstruction result based on GPS, PALSAR, and E-CHAIM model with 10 and 30 min, respectively.

For example, when the true FRA is approximately equal to $n\pi/4$, Freeman's second estimator will cause a large deviation due to the dominant role of noise [31], and Chen and Quegan's [25] fourth to sixth estimators will fail in areas where the reflection symmetry assumptions are satisfied, such as forest area.

D. Effects of Voxel Resolution on CIT

The construction of the voxel model is also an important component of CIT. It provides the basis for establishing and calculating subsequent tomographic equations. The chosen

voxel size will unavoidably affect the CIT's quality and precision. Therefore, this section explores the effect of different voxel sizes on the CIT accuracy. Compared to the extensive research on inversion algorithms, voxel construction in CIT has received relatively little attention.

The results (AD and RMSE) of 11 groups that have different voxel numbers (i.e., voxel resolution) are presented in Tables I–IV, the other parameters of CIT are the same as Figs. 2 and 4. The results include the voxel number, AD, RMSE, and the change rate before and after tomography (i.e., the ratio of tomography results to initial value). A lower rate of change indicates a greater tomographic effect. From

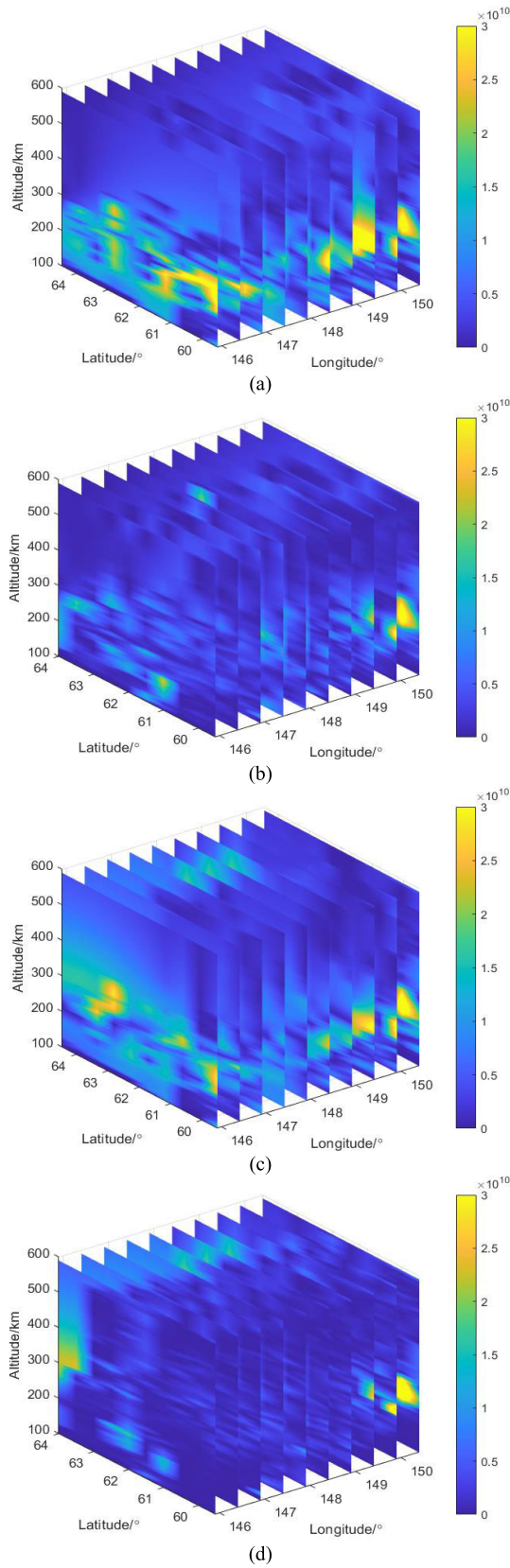


Fig. 5. Deviations of CIT of Fig. 4. (a) and (b) Deviations between the true distribution and reconstructions based on GPS and E-CHAIM with 10 and 30 min, respectively. (c) and (d) Deviations between the true distribution and reconstructions based on GPS, PALSAR, and E-CHAIM with 10 and 30 min, respectively.

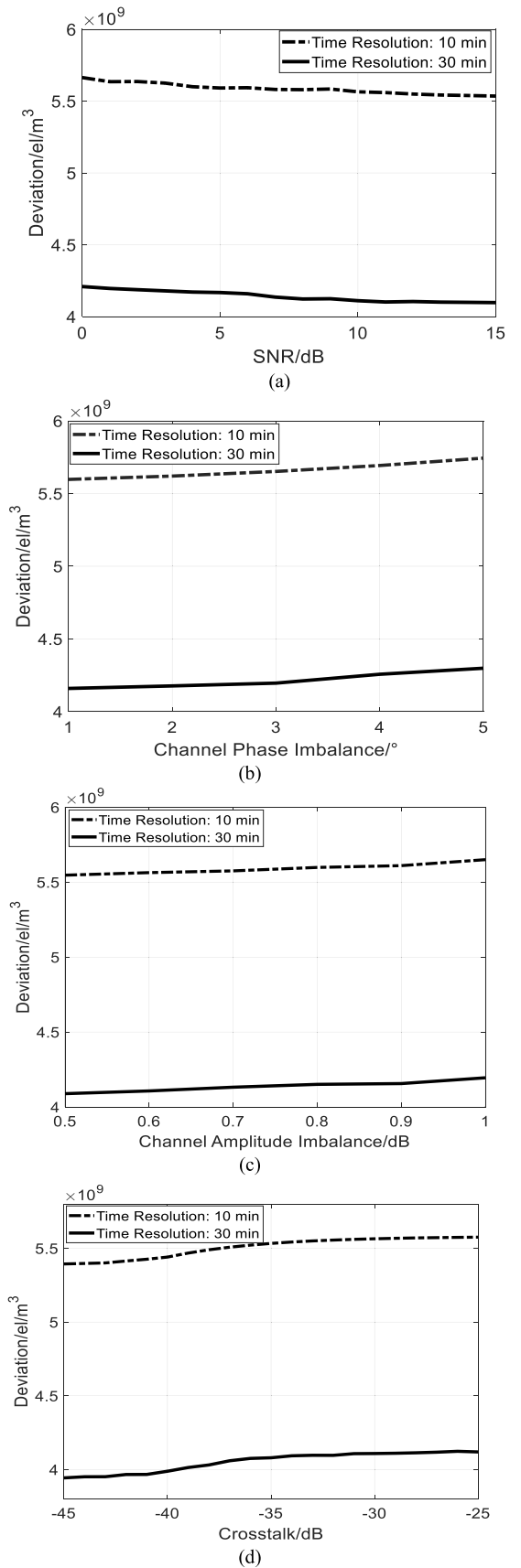


Fig. 6. ADs of the proposed CIT with (a) SNR, (b) channel phase imbalance, (c) channel amplitude imbalance, and (d) crosstalk. Other conditions are as for Figs. 2 and 4.

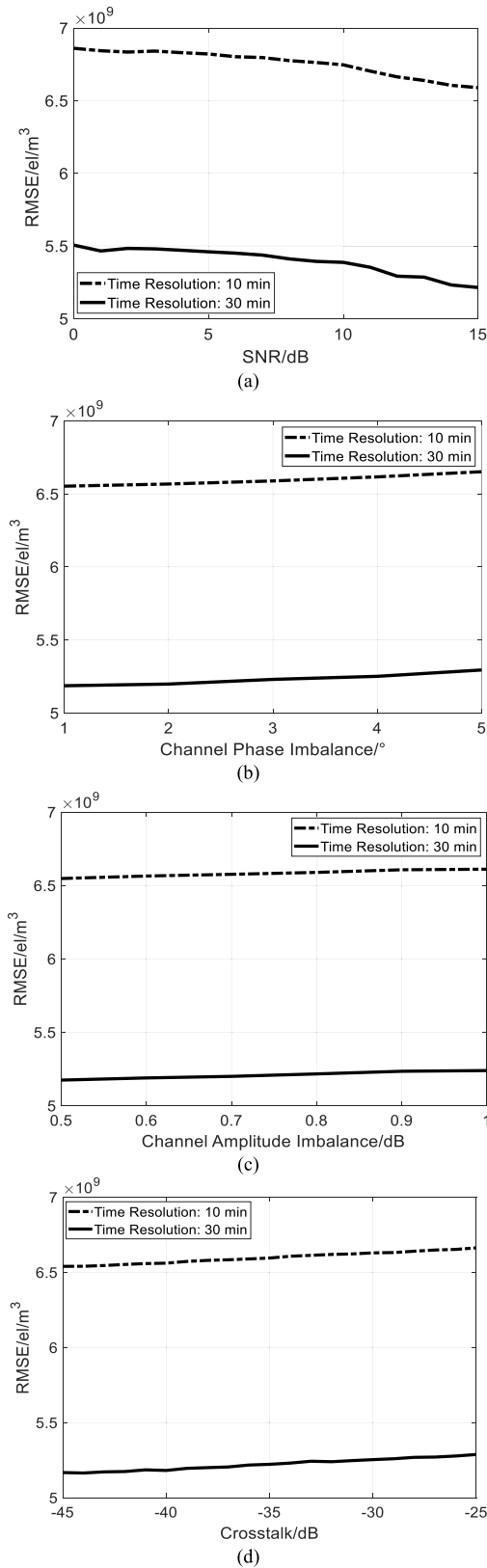


Fig. 7. RMSE of the proposed CIT with (a) SNR, (b) channel phase imbalance, (c) channel amplitude imbalance, and (d) crosstalk. Other conditions are as for Figs. 2 and 4.

the results, the fewer the voxel numbers, the smaller the change. This may indicate a more significant tomographic effect, but it does not necessarily mean a higher quality of tomographic imaging. This is due to the fact that accuracy

TABLE I
AVERAGE DEVIATION OF DIFFERENT VOXEL SIZES WITH 10 MIN

Voxel numbers ($lat \times lon \times alt$)	AD of initial value (el/m^3)	AD of CIT (el/m^3)	Change rate (%)
$2 \times 2 \times 10$	6.85×10^9	1.52×10^9	22.2
$2 \times 2 \times 25$	6.51×10^9	1.60×10^9	24.6
$2 \times 2 \times 50$	6.41×10^9	1.71×10^9	26.6
$3 \times 3 \times 50$	6.50×10^9	2.78×10^9	42.7
$4 \times 4 \times 50$	6.58×10^9	3.64×10^9	55.3
$5 \times 5 \times 50$	6.62×10^9	4.44×10^9	67.2
$10 \times 10 \times 50$	6.68×10^9	5.52×10^9	83.0
$20 \times 20 \times 50$	6.72×10^9	6.34×10^9	94.1
$20 \times 50 \times 50$	6.76×10^9	6.46×10^9	95.6
$20 \times 50 \times 25$	6.83×10^9	6.45×10^9	94.5
$20 \times 50 \times 10$	7.10×10^9	6.51×10^9	91.7

TABLE II
AD OF DIFFERENT VOXEL SIZES WITH 30 MIN

Voxel numbers ($lat \times lon \times alt$)	AD of initial value (el/m^3)	AD of CIT (el/m^3)	Change rate (%)
$2 \times 2 \times 10$	6.85×10^9	0.26×10^9	3.8
$2 \times 2 \times 25$	6.51×10^9	0.20×10^9	3.1
$2 \times 2 \times 50$	6.41×10^9	0.18×10^9	2.8
$3 \times 3 \times 50$	6.50×10^9	0.70×10^9	10.8
$4 \times 4 \times 50$	6.58×10^9	1.41×10^9	21.4
$5 \times 5 \times 50$	6.62×10^9	2.52×10^9	38.1
$10 \times 10 \times 50$	6.68×10^9	4.56×10^9	68.3
$20 \times 20 \times 50$	6.72×10^9	5.38×10^9	79.6
$20 \times 50 \times 50$	6.76×10^9	5.92×10^9	87.6
$20 \times 50 \times 25$	6.83×10^9	6.01×10^9	87.9
$20 \times 50 \times 10$	7.10×10^9	6.17×10^9	86.9

and voxel resolution are two distinct evaluation criteria for CIT, and only when both are high is the tomographic image quality improved. Even though the resolution improves as the number of voxel increases, this inevitably leads to a rise in the number of unknowns in the tomographic equation, which affects the equation's stability. In conditions where the number of rays remains constant, increasing the voxel number may

TABLE III
RMSE OF DIFFERENT VOXEL SIZES WITH 10 MIN

Voxel numbers ($lat \times lon \times alt$)	RMSE of initial value (el/m^3)	RMSE of CIT (el/m^3)	Change rate (%)
$2 \times 2 \times 10$	7.01×10^9	1.53×10^9	21.8
$2 \times 2 \times 25$	6.99×10^9	1.76×10^9	25.2
$2 \times 2 \times 50$	6.93×10^9	1.77×10^9	25.5
$3 \times 3 \times 50$	7.15×10^9	2.71×10^9	37.9
$4 \times 4 \times 50$	7.23×10^9	3.88×10^9	53.7
$5 \times 5 \times 50$	7.29×10^9	4.71×10^9	64.6
$10 \times 10 \times 50$	7.42×10^9	6.54×10^9	88.1
$20 \times 20 \times 50$	7.48×10^9	7.12×10^9	95.2
$20 \times 50 \times 50$	7.49×10^9	7.23×10^9	96.5
$20 \times 50 \times 25$	7.53×10^9	7.24×10^9	96.2
$20 \times 50 \times 10$	7.58×10^9	7.16×10^9	94.5

TABLE IV
RMSE OF DIFFERENT VOXEL SIZES WITH 30 MIN

Voxel numbers ($lat \times lon \times alt$)	RMSE of initial value (el/m^3)	RMSE of CIT (el/m^3)	Change rate (%)
$2 \times 2 \times 10$	7.01×10^9	0.67×10^9	9.6
$2 \times 2 \times 25$	6.99×10^9	0.51×10^9	7.3
$2 \times 2 \times 50$	6.93×10^9	0.39×10^9	5.8
$3 \times 3 \times 50$	7.15×10^9	1.10×10^9	15.4
$4 \times 4 \times 50$	7.23×10^9	2.26×10^9	31.2
$5 \times 5 \times 50$	7.29×10^9	3.03×10^9	41.6
$10 \times 10 \times 50$	7.42×10^9	5.69×10^9	76.7
$20 \times 20 \times 50$	7.48×10^9	6.26×10^9	83.7
$20 \times 50 \times 50$	7.49×10^9	6.64×10^9	88.7
$20 \times 50 \times 25$	7.53×10^9	6.69×10^9	88.8
$20 \times 50 \times 10$	7.58×10^9	6.71×10^9	88.5

actually decrease the accuracy of tomographic imaging. This is consistent with the conclusion of [53]. Although reducing the voxel resolution will allow more rays to pass through each voxel, thereby improving the CIT accuracy, it also means that the area corresponding to each voxel will be enlarged, and the electron density value in the voxel is generally assumed to be a constant, that is, the same value is used to represent the true electron density distribution of a larger area. However, the true

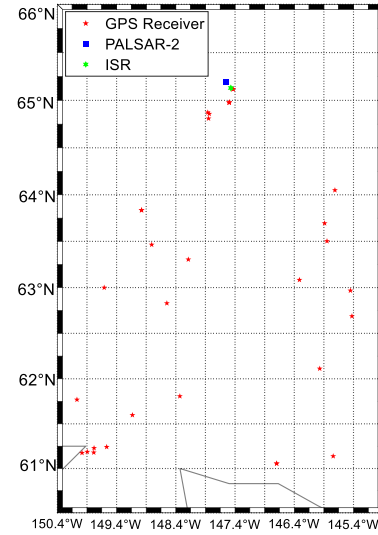


Fig. 8. Geographic relationship of GPS receiver, PALSAR-2, and ISR.

electron density is a spatially variable region. As illustrated in Fig. 2(a), even adjacent electron density is not identical. This distinction is most evident in the region of the F2 layer with the highest electron density. Consequently, a decrease in voxel resolution will increase the ambiguity of representing the true distribution with corresponding values. In turn, this impacts the quality of CIT from another perspective.

Consequently, the first stage of CIT is to evaluate the voxel resolution and change rate. According to the tables, the effect of different vertical resolutions on the rate of change is not significant. However, when the numbers of horizontal voxels change from 10×10 to 2×2 , there is a noticeable improvement in the rate of change, whereas the rate of change does not change considerably when the number of horizontal voxels exceeds 10×10 .

Since the horizontal range of the reconstructed region in this article is $5^\circ \times 5^\circ$, the optimal horizontal resolution for voxel sizes of 10×10 to 2×2 is $0.5^\circ \times 0.5^\circ$ to $2.5^\circ \times 2.5^\circ$. This also confirms why the majority of GPS-based tomographic imaging utilizes voxel sizes within this range [38], [39], [40], [41], [42], [43].

E. Verification Using Measured Data

In this section, we utilize a set of incoherent scatter radar (ISR) data near the Poker Flat, Alaska, PALSAR-2 data, E-CHAIM model, and GPS data with 10- and 30-min resolutions for experimental verification.

As shown in Fig. 8, the reconstruction area is established within a latitude range of 61°N – 66°N and a longitude range of 145.4°W – 150.4°W , at an altitude ranging from 100 to 600 km. The green hexagon represents the location of the ISR (147.47°W , 65.13°N). The electron density profile obtained by ISR is used as the true value for comparison, and is obtained at 22:20 UT, on August 29, 2014. The blue square nearby is the PALSAR-2 observation point, and its observation time is almost identical to the ISR, at 22:24 UT, on August 29, 2014. Different from the simulation data, there is currently only one

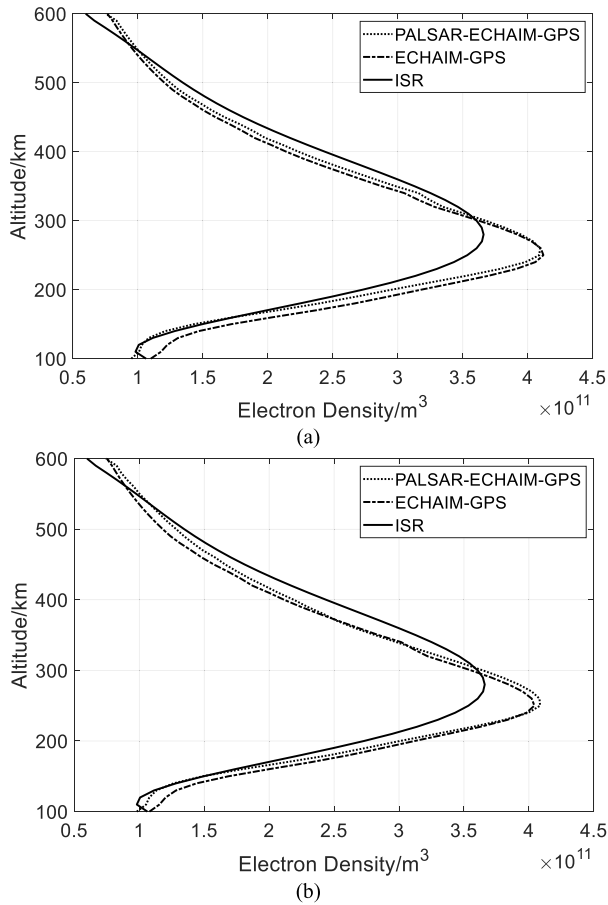


Fig. 9. CIT results at the position of ISR with (a) 10 and (b) 30 min.

PALSAR-2 measured data that meets the CIT requirements, so a reasonable assumption needs to be made, that is, the ionosphere is relatively quiet during the PALSAR observation, and the ionospheric distribution in the reconstruction area can be considered the same as that at the PALSAR-2 observation point. In this case, the vertical TEC retrieved by PALSAR-2 is 11.4 TECU, and the ISR retrieval is 12.2 TECU, indicating that the PALSAR measured data can reflect the real ionospheric conditions. By using the TEC values obtained from PALSAR-2 and (14), the initial electron density values obtained from the E-CHAIM model within the reconstruction area can be uniformly corrected to obtain the corrected initial values for iteration. The red pentagrams in Fig. 8 are the ground GPS receiving stations. The GPS data obtained are consistent with the ISR and PALSAR-2 in time [50]. Two time resolutions between 22:16 and 22:25 UT on August 29, 2014 (i.e., 10 min) and between 22:06 and 22:35 UT on August 29, 2014 (i.e., 30 min) are selected. In the third step of CIT, the measured TEC_i value of each ray (i.e., the matrix $\mathbf{d}$) can be read from the GPS data (los_20140829.001.h5 file). Other parameters and CIT process are the same as the simulations.

Fig. 9 shows the comparison between the electron density profile obtained by the ISR and the results based on the GPS-ECHAIM and GPS-ECHAIM-PALSAR algorithms. Fig. 9(a) and (b) are the comparison results of the electron density profiles with 10-and 30-min time resolutions,

respectively. In Fig. 9(a), the AD and RMSE between the reconstruction results of the model (i.e., ECHAIM-GPS) and the true value (i.e., ISR) are the $2.46 \times 10^{10} \text{ el/m}^3$ and $2.91 \times 10^{10} \text{ el/m}^3$, respectively. On the other hand, for the proposed method (i.e., GPS-ECHAIM-PALSAR), the AD and RMSE between the reconstruction and the true value are $2.14 \times 10^{10} \text{ el/m}^3$ and $2.31 \times 10^{10} \text{ el/m}^3$, respectively. This represents an improvement of 13.01% and 20.62% compared to the accuracy of the model alone. Similarly, in Fig. 9(b), the AD and RMSE between the reconstruction results of the model alone and the true value are $2.29 \times 10^{10} \text{ el/m}^3$ and $2.74 \times 10^{10} \text{ el/m}^3$, respectively, while the AD and RMSE between the reconstruction results of the proposed method and the true value are $2.05 \times 10^{10} \text{ el/m}^3$ and $2.19 \times 10^{10} \text{ el/m}^3$, respectively, with accuracy 10.48% and 20.1% higher than that of the model alone, respectively. It can be seen that the proposed method, which incorporates PALSAR data correction into the initial value, achieves better overall similarity to the ISR results, demonstrating an enhancement in the accuracy of tomographic imaging.

IV. SUMMARY AND CONCLUSION

The CIT is currently the main technique used for 3-D reconstruction of the ionospheric structure. Starting from the perspective of initial values, this article modifies the E-CHAIM model with fully polarized PALSAR data to derive initial value distributions that are closer to the actual conditions, thereby improving the tomographic precision. Numerical experiments show that the AD and RMSE decrease by at least 16% and 30%, respectively, using the corrected initial values for tomography during the quiet ionosphere at two different time resolutions. During period of disturbance, the AD and RMSE decreased by at least 14% and 20%, respectively. Furthermore, the measured data verification in Alaska shows that the AD and RMSE of the proposed method decrease by at least 10% and 20%, respectively, compared with the model method only, which verifies the effectiveness of the proposed method. In addition, this study has evaluated the effect of different PALSAR system errors on tomographic imaging. The results indicate that the algorithm proposed in this article exhibits good robustness against system errors. The effects of different voxel resolutions on CIT are then analyzed. In order to improve the quality of tomographic imaging, the results demonstrate that in addition to optimizing the algorithm, voxel division should be carefully considered.

In general, due to the combination of all-day, all-weather, and two-way propagation characteristics of satellite SAR, the CIT proposed in this article, which combines full-polarized PALSAR and empirical model data, has obvious advantages in feasibility, robustness, and easy implementation, and effectively improve the tomography accuracy. In the future, the improvement of the CIT efficiency of the variable grid scale and the verification by using more measured data need to be studied.

ACKNOWLEDGMENT

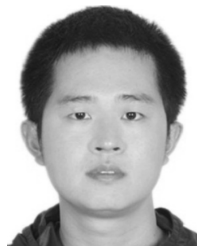
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Cheng Wang was born in Shaanxi, China, in 1988. He received the B.S. degree in applied physics and the Ph.D. degree in radio physics from Xidian University, Xi'an, China, in 2010 and 2015, respectively.

He is currently an Associate Professor with the Qian Xuesen Laboratory of Space Technology, China Academy of Space Technology, Beijing, China. His research interests include ionospheric radio propagation and ionospheric sounding.

Hai-Sheng Zhao, photograph and biography not available at the time of publication.

Li-Ming Wang, photograph and biography not available at the time of publication.

Le Cao (Member, IEEE), photograph and biography not available at the time of publication.

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